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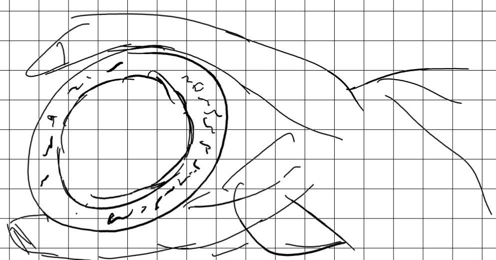
AN INEQUALITY TO RULE THEM ALL

Potenzfkt ≪ Exponentials ≪ Factorials



an inequality to rule them all

$$\boxed{\text{power functions}} \underset{①}{\ll} \boxed{\text{exponential functions}} \underset{②}{\ll} \boxed{\text{factorials}} \underset{③}{\ll} \boxed{\text{super-exponentials}}$$

① for all $a>1$, $r \geq 0$ we have $\lim \frac{n^r}{a^n} = 0$.

Take $m \in \mathbb{N}$ such that $m \geq r$, $b>0$ such that $a = (1+b)^m$.

$$0 \leq \frac{n^r}{a^n} \leq \frac{n^m}{(1+b)^{mn}} = \left(\frac{n}{(1+b)^n}\right)^m = \left(\frac{n}{1+nb+\binom{n}{2}b^2 + \text{more}}\right)^m$$

$$\leq \left(\frac{n}{\frac{n(n-1)}{2}b^2}\right)^m = \left(\frac{2}{b^2(n-1)}\right)^m \xrightarrow[n\to\infty]{} 0$$

So ① by the squeezing-theorem. □

② For all $a>1$, we have $\lim\limits_{n\to\infty} \frac{a^n}{n!} = 0$.

Find $m \in \mathbb{N}$ such that $m \geq a$.

$$0 \leq \frac{a^n}{n!} \leq \frac{m^n}{n!} \underset{=}{\leq} \frac{m^{m+k}}{(m+k)!} = \left(\frac{m^m}{m!}\right) \cdot \left(\frac{m^k}{(m+1) \cdots (m+k)}\right)$$

only when $n > m$, but ok cause $n \to \infty$
$\Rightarrow n = m+k$ for $k \in \mathbb{N}$

$$= c \cdot \frac{m}{m+1} \cdot \frac{m}{m+2} \cdot \ldots \cdot \frac{m}{m+k} \leq c \underbrace{\left(\frac{m}{m+1}\right)}_{0< \quad <1}{}^k \xrightarrow{k\to\infty} 0$$

□

③ $\lim\limits_{n\to\infty} \frac{n!}{n^n} = 0$

$$0 \le \frac{n!}{n^n} = \frac{1\cdot 2\cdot 3\cdot \ldots \cdot (n-1)\cdot \cancel{n}}{n\cdot n\cdot n\cdot \ldots \cdot n\cdot \cancel{n}} \le \frac{1}{n} \xrightarrow{n\to\infty} 0$$

$0 < \ldots < 1$

$\square$